Network Security Implementation
Laboratory Report

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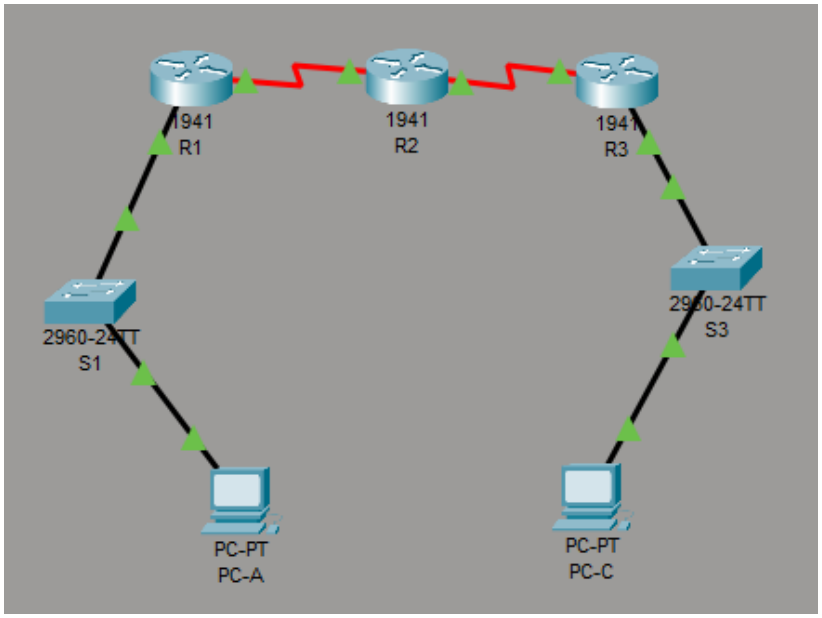
Task 1

1. Introduction

This lab focuses on configuring a basic network using Cisco Packet Tracer. The objective is to connect routers, switches, and PCs, assign IP addresses, configure OSPF routing, and verify connectivity between devices.

2. Network Topology

The network consists of three routers (R1, R2, R3), two switches (S1, S3), and two PCs (PC-A and PC-C). The devices are connected in a linear topology: PC-A → S1 → R1 → R2 → R3 → S3 → PC-C.



3. IP Address Configuration

Device	Interface	IP Address
PC-A	FastEthernet0	192.168.1.2
R1	Gig0/0	192.168.1.1
R1	Serial	10.1.1.1
R2	Serial	10.1.1.2 / 10.2.2.1
R3	Serial	10.2.2.2
R3	Gig0/0	192.168.3.1
PC-C	FastEthernet0	192.168.3.2

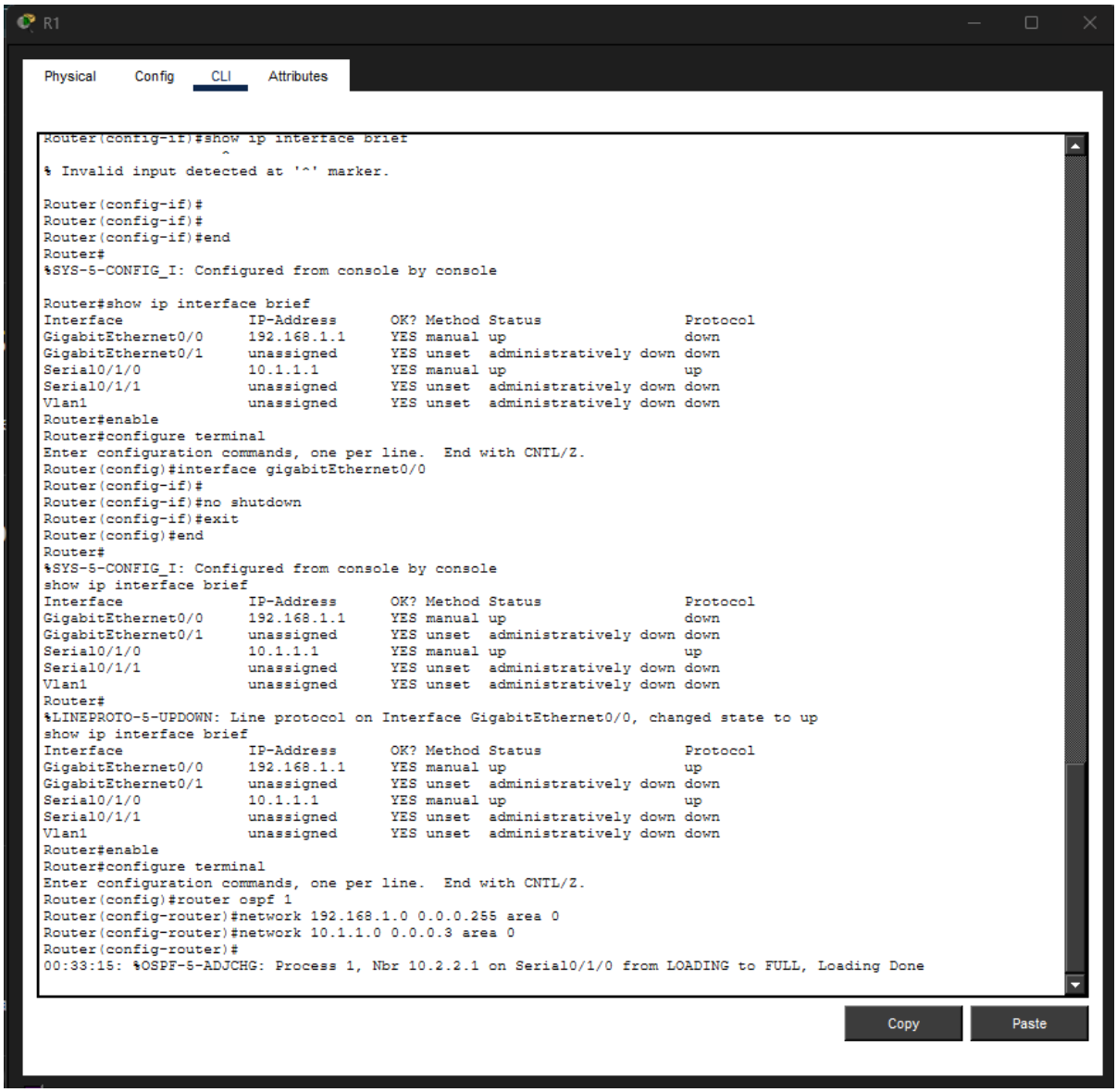
```
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.1        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#enable
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.2        YES manual up                    up
Serial0/1/1            10.2.2.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.3.1     YES manual up                    down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.2.2.2        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

4. Router Configuration

Routers were configured using CLI commands to assign IP addresses, enable interfaces using 'no shutdown', and configure serial connections.




```
Router(config-if)#show ip interface brief
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.1        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.1        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.1        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#
00:33:15: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done
```

R2

Physical | Config | CLI | Attributes

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.2        YES manual up                    up
Serial0/1/1            10.2.2.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#





Router con0 is now available




Press RETURN to get started.


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1'
                             ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
00:32:48: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/1/0 from LOADING to FULL, Loading Done

00:33:18: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/1/1 from LOADING to FULL, Loading Done
```

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R3

Physical Config CLI Attributes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface serial0/1/0
Router(config-if)#ip address 10.2.2.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#Router#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router#Interface              IP-Address      OK?
% Unrecognized command
Router#Interface              IP-Address      OK Method Status                Protocol
                              ^
% Invalid input detected at '^' marker.

Router#GigabitEthernet0/0     unassigned      YES unset  administratively down down
                              ^
% Invalid input detected at '^' marker.

Router#GigabitEthernet0/1     unassigned      YES unset  administratively down down
                              ^
% Invalid input detected at '^' marker.

Router#Serial0/1/0            10.1.1.2        YES manual up                    up
                              ^
% Invalid input detected at '^' marker.

Router#Serial0/1/1            10.2.2.1        YES manual up                    up
                              ^
% Invalid input detected at '^' marker.

Router#Vlan1                  unassigned      YES unset  administratively down down^Z
```

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5. OSPF Configuration

OSPF routing protocol was configured on all routers to enable communication between networks. R1 advertises 192.168.1.0 and 10.1.1.0, R2 advertises 10.1.1.0 and 10.2.2.0, and R3 advertises 192.168.3.0 and 10.2.2.0.

```
Router>enable
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
00:21:27: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done
```

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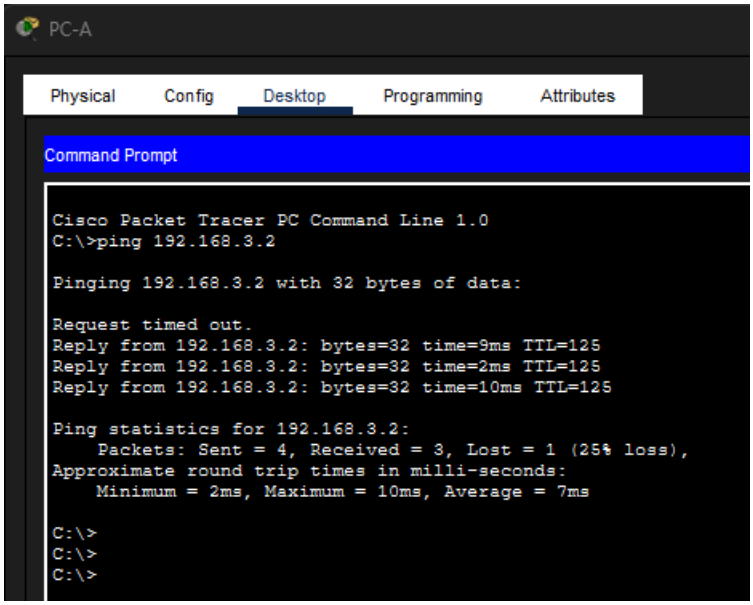
6. Connectivity Testing

Connectivity was verified using the ping command from PC-A to PC-C. Successful replies were received, confirming proper routing. Initial packet loss is normal due to ARP.



7. Conclusion

The network was successfully configured and tested. All devices were able to communicate using OSPF routing. This lab demonstrated basic network setup, IP addressing, and routing configuration.

Task 2

1. Introduction

This task focuses on securing network devices by implementing basic security configurations on routers. The objective is to protect access using passwords, encrypt credentials, display warning banners, and enable secure remote access using SSH.

2. Password Configuration

Passwords were configured on all routers to prevent unauthorized access. The enable secret command was used to secure privileged mode, while console and VTY passwords were configured for local and remote access.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret class
Router(config)#
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#
Router(config)#service password-encryption
Router(config)#banner motd #WARNING: Unauthorized access is prohibited#
Router(config)#username admin secret admin123
Router(config)#ip domain-name vit.local
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#hostname R1
R1(config)#crypto key generate rsa
The name for the keys will be: R1.vit.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#line vty 0 4
*Mar 1 0:3:40.135: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
R1(config)#
```

3. Password Encryption

The service password-encryption command was used to encrypt all configured passwords, ensuring they are not stored in plain text.

4. Banner Configuration

A Message of the Day (MOTD) banner was configured to warn users that unauthorized access is prohibited. This banner is displayed before login.

```
C:\>
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=6ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ssh -l admin 192.168.1.1

Password:

WARNING: Unauthorized access is prohibited

R1>
```

5. SSH Configuration

Secure Shell (SSH) was configured to allow secure remote access to the routers. A hostname and domain name were set, RSA keys were generated, and a local user account was created. VTY lines were configured to accept only SSH connections.

R2

Physical | Config | CLI | Attributes

```
down or detached

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

01:27:32: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/1/0 from LOADING to FULL, Loading Done


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
R2(config)#enable secret class
R2(config)#service password-encryption
R2(config)#
R2(config)#line console 0
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#exit
R2(config)#
R2(config)#line vty 0 4
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#exit
R2(config)#
R2(config)#banner motd #WARNING: Unauthorized access is prohibited#
R2(config)#
R2(config)#username admin secret admin123
R2(config)#
R2(config)#ip domain-name vit.local
R2(config)#crypto key generate rsa
The name for the keys will be: R2.vit.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R2(config)#line vty 0 4
*Mar 1 1:33:42.340: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2(config-line)#transport input ssh
R2(config-line)#login local
R2(config-line)#exit
R2(config)#
R2(config)#ip ssh version 2
R2(config)#end
R2#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

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R3

Physical Config CLI Attributes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#
R3(config)#enable secret class
R3(config)#service password-encryption
R3(config)#
R3(config)#line console 0
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#exit
R3(config)#
R3(config)#line vty 0 4
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#exit
R3(config)#
R3(config)#banner motd #WARNING: Unauthorized access is prohibited#
R3(config)#
R3(config)#username admin secret admin123
R3(config)#
R3(config)#ip domain-name vit.local
R3(config)#crypto key generate rsa
The name for the keys will be: R3.vit.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#line vty 0 4
*Mar 1 1:23:7.964: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config-line)#transport input ssh
R3(config-line)#login local
R3(config-line)#exit
R3(config)#
R3(config)#ip ssh version 2
R3(config)#end
R3#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

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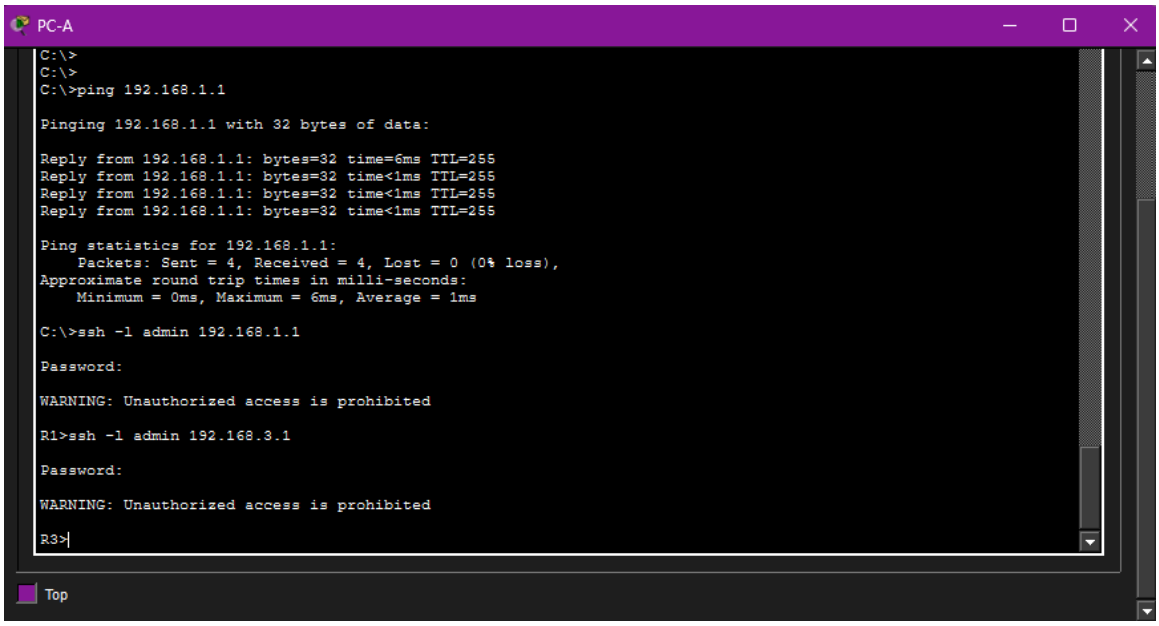
6. Testing and Verification

SSH connectivity was tested from a PC using the ssh command. Successful login confirmed that secure remote access was properly configured. The warning banner was also displayed during login.



7. Conclusion

Router security was successfully implemented using passwords, encryption, banners, and SSH. These configurations enhance network security by preventing unauthorized access and ensuring secure communication.

Task 3

1. Introduction

This task focuses on implementing a Virtual Private Network (VPN) using IPSec between two routers. The purpose is to provide secure communication between two networks by encrypting transmitted data.

2. Objective

- Configure IPSec VPN between Router R1 and Router R3
- Secure communication between PC-A and PC-C
- Ensure data confidentiality using encryption

3. IPSec VPN Configuration

3.1 ISAKMP Policy (Phase 1)

```
crypto isakmp policy 10
 encryption aes
 hash sha
 authentication pre-share
 group 2
 lifetime 86400
```

3.2 Pre-shared Key

```
crypto isakmp key cisco address 10.2.2.2
```

3.3 IPSec Transform Set (Phase 2)

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
```

3.4 Access Control List (ACL)

```
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

3.5 Crypto Map

```
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.2.2.2
 set transform-set VPN-SET
 match address 100
```

3.6 Applying Crypto Map

```
interface serial0/1/0
 crypto map VPN-MAP
```

4. Testing and Verification

The VPN connection is verified by sending ping packets between PC-A and PC-C.
ping 192.168.3.2

Successful communication indicates that the VPN tunnel is functioning correctly.

5. Limitation in Cisco Packet Tracer

During this task, IPSec VPN configuration could not be fully implemented in Cisco Packet Tracer. This is because Packet Tracer has limited support for advanced security features such as IPSec. Commands like 'crypto isakmp policy' are not recognized in some router models (e.g., Cisco 1941) due to missing security licenses or modules. Therefore, the configuration is demonstrated for theoretical understanding rather than full practical implementation.

6. Conclusion

The IPSec VPN configuration demonstrates how secure communication can be established between networks. Although full implementation was not possible in Packet Tracer, the concepts of encryption, authentication, and secure tunneling were successfully understood.

Task 4

Introduction

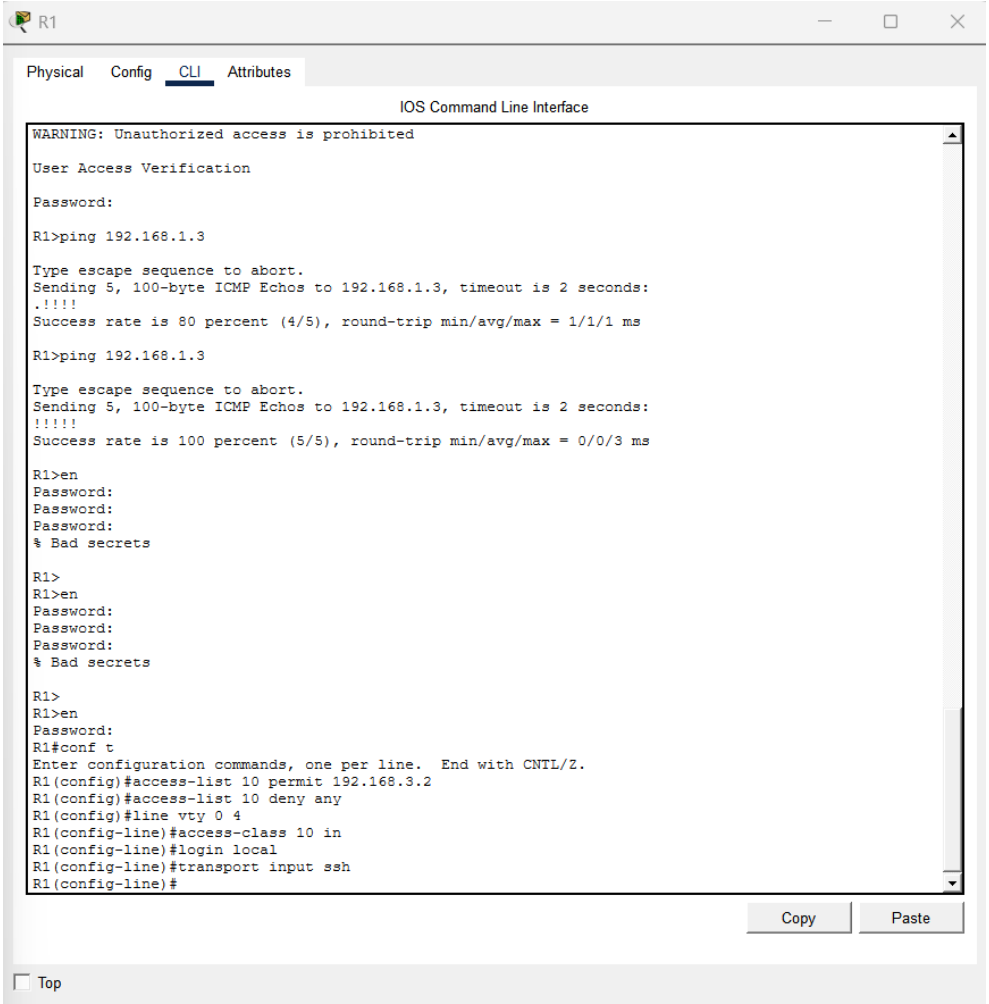
This task focuses on controlling administrative access to network devices using Access Control Lists (ACLs). The objective is to ensure that only the designated management PC (PC-C) is allowed to access routers R1, R2, and R3, while all other devices are restricted.

Objective

- Restrict router access to PC-C only
- Prevent unauthorized access from other hosts
- Verify access control using SSH connectivity

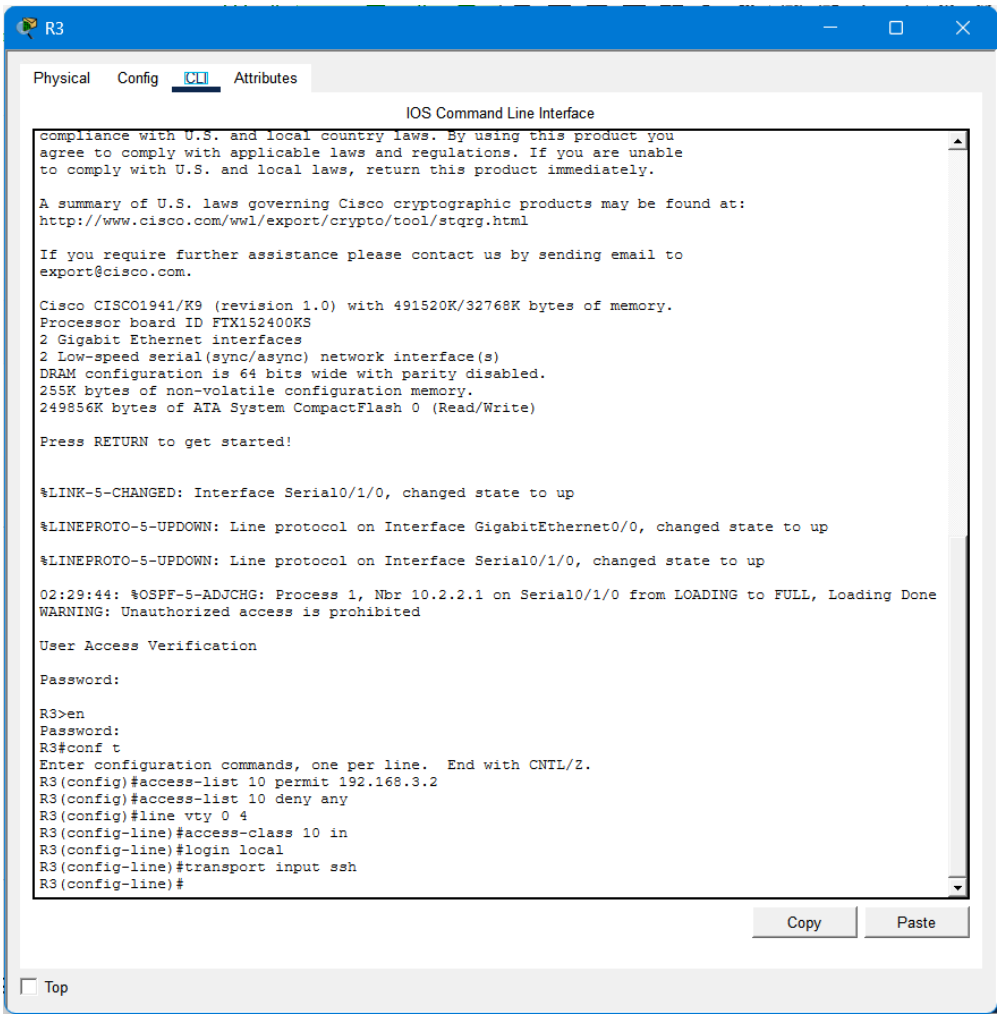
ACL Configuration

A standard ACL was configured on routers R1 and R3 to permit access only from PC-C (192.168.3.2) and deny all other hosts.



Configuration commands:

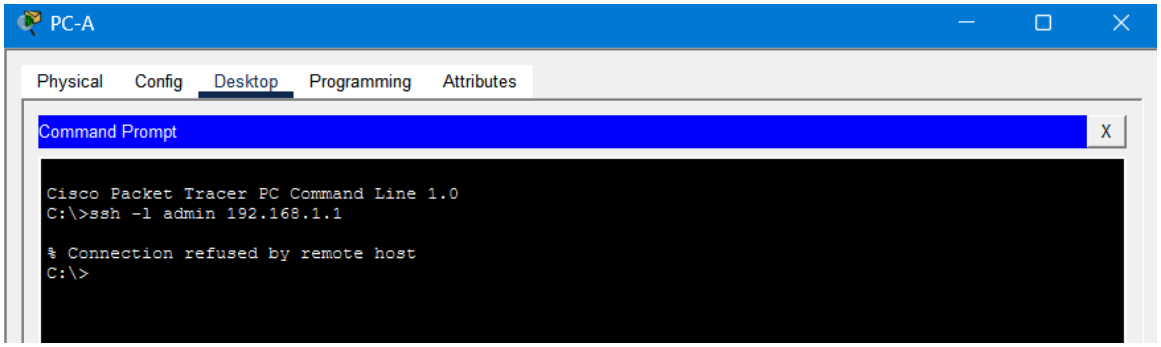
access-list 10 permit 192.168.3.2
access-list 10 deny any
line vty 0 4
access-class 10 in
login local
transport input ssh

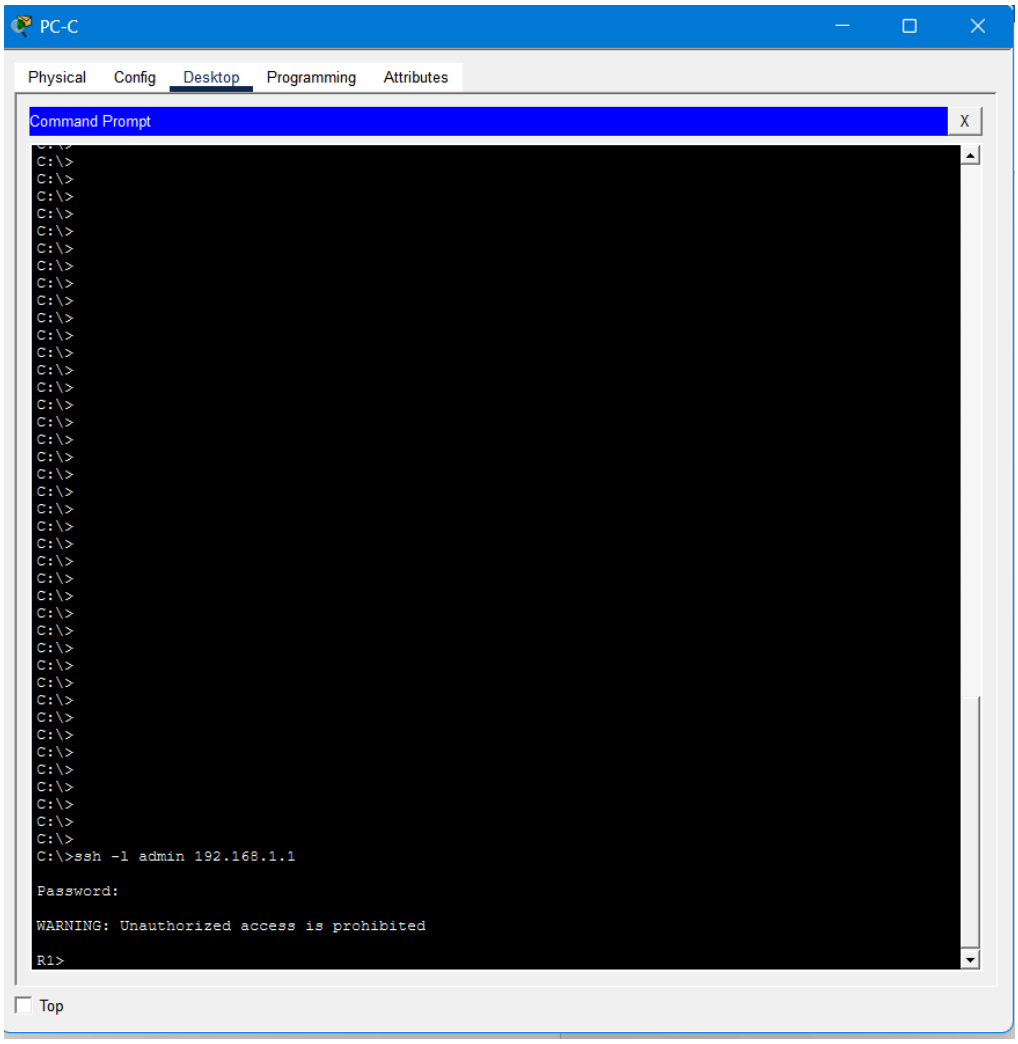


Testing and Verification

- From **PC-C**, SSH access to the routers was successful.
- From **PC-A**, SSH access was denied with a connection refused message.

This confirms that only authorized devices can access the routers





Conclusion

The ACL was successfully implemented to control administrative access. Unauthorized access attempts were blocked, enhancing network security.

Task 5

Introduction

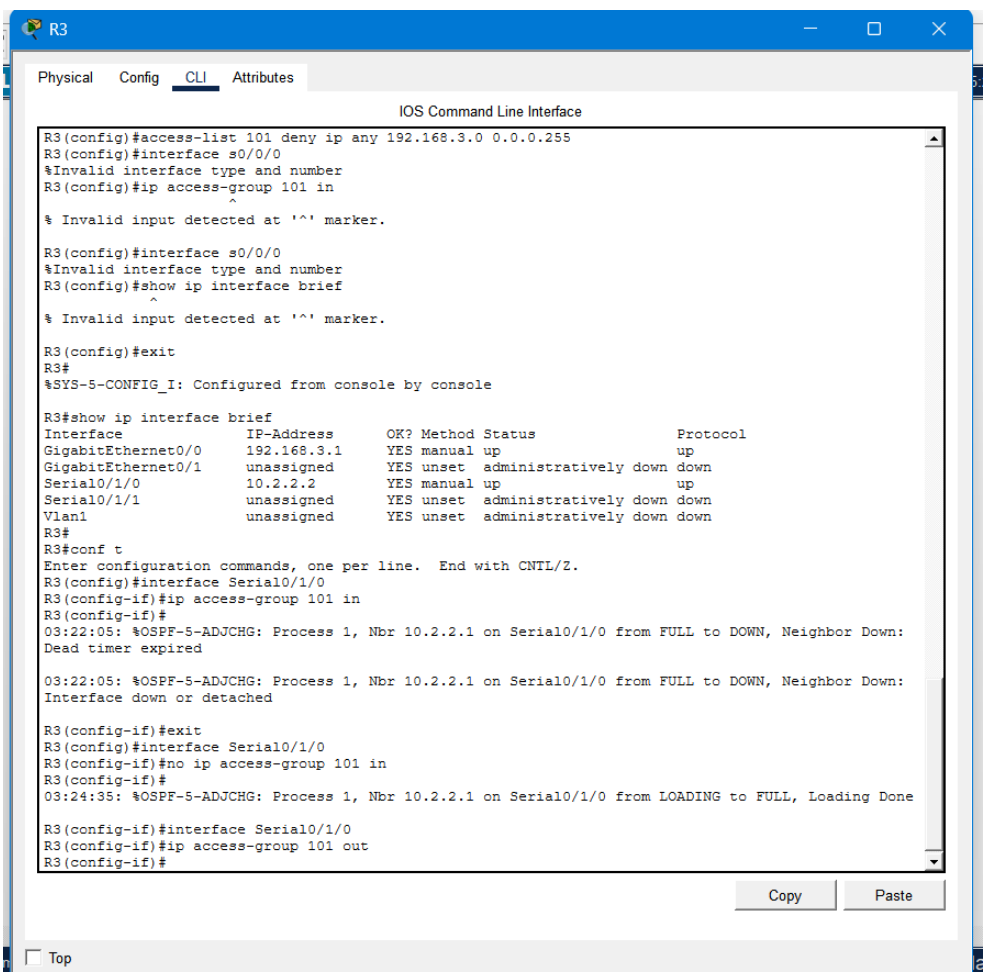
This task involves implementing firewall functionality on router R3 to control traffic flow between internal and external networks. Due to limitations of Cisco Packet Tracer, Zone-Based Firewall (ZPF) was not supported; therefore, an extended ACL was used to simulate firewall behavior.

Objective

- Allow internal network traffic to access external resources
- Block unauthorized external traffic from accessing internal network
- Verify traffic filtering functionality

Firewall Configuration (Using ACL)

An extended ACL was configured on R3 to permit traffic from the internal network (192.168.3.0/24) and restrict unauthorized inbound traffic.

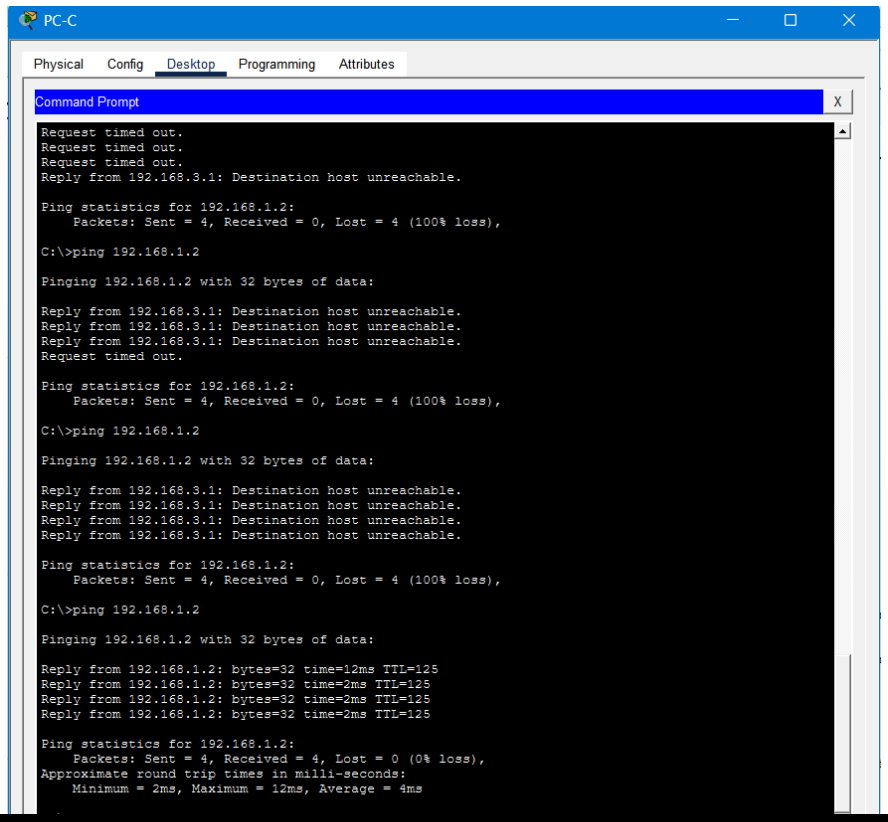


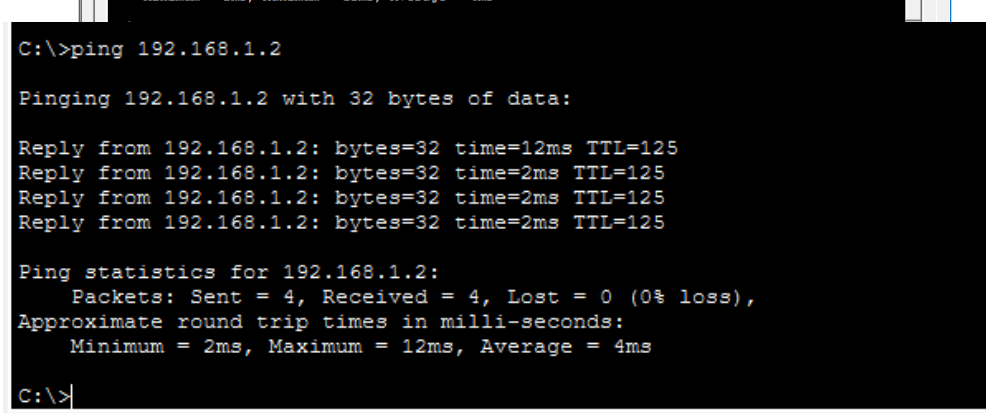
Configuration commands:

```
access-list 101 permit ip 192.168.3.0 0.0.0.255 any
access-list 101 deny ip any 192.168.3.0 0.0.0.255
interface Serial0/1/0
ip access-group 101 out
```

Testing and Verification

- Internal host (PC-C) successfully communicated with PC-A.
- Traffic from external sources to the internal network was restricted.
- Ping tests confirmed correct traffic flow and filtering behavior.





Conclusion

Firewall functionality was successfully implemented using ACLs. The configuration ensured that internal users could access external resources while protecting the internal network from unauthorized access.

Task 6

Introduction

This task focuses on implementing network monitoring using Syslog to simulate intrusion detection functionality. Due to limitations of Cisco Packet Tracer, full IPS configuration was not supported; therefore, logging and monitoring mechanisms were used instead.

Objective

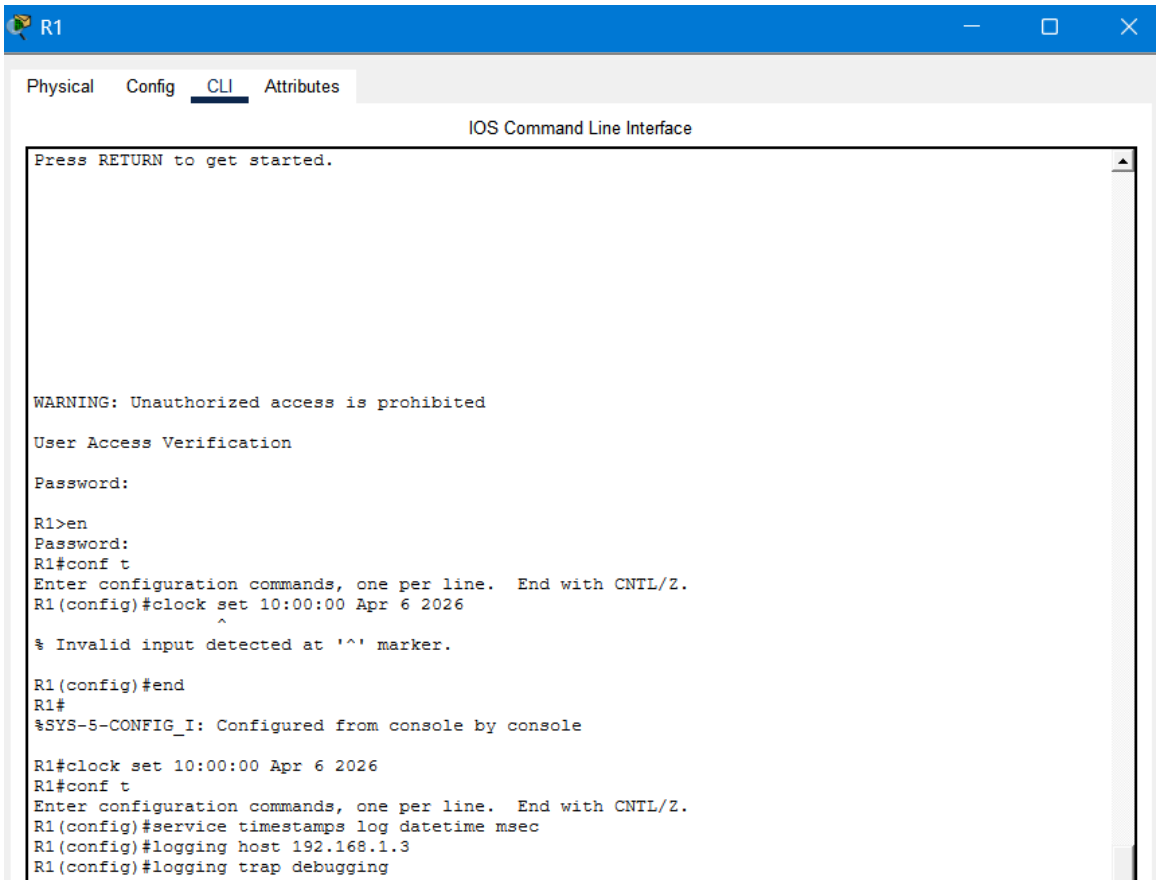
- Enable logging of network events
- Configure a Syslog server to receive logs
- Monitor network activity for security analysis

Configuration

Time Configuration

clock set 10:00:00 Apr 6 2026

service timestamps log datetime msec



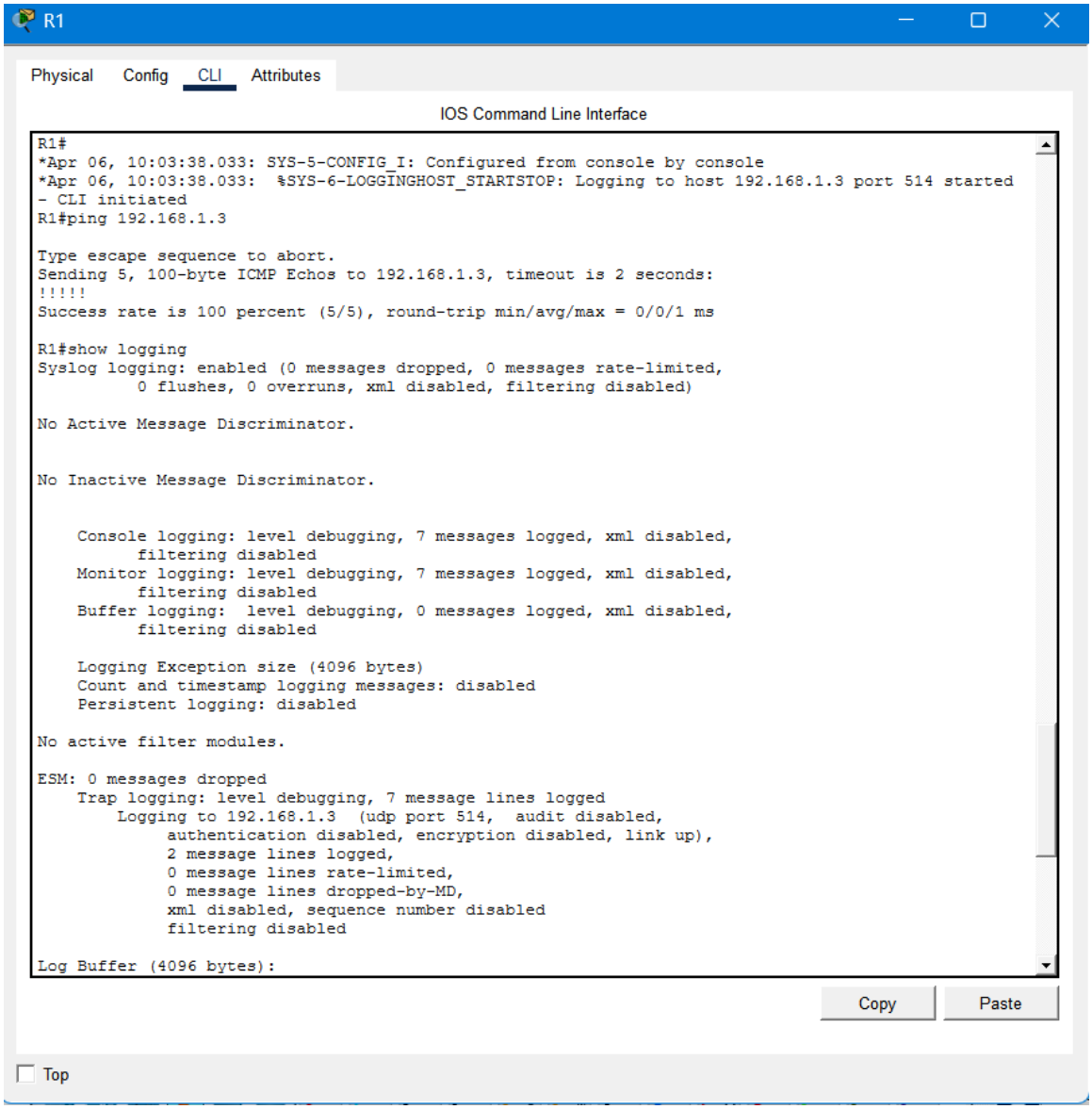
Syslog Configuration

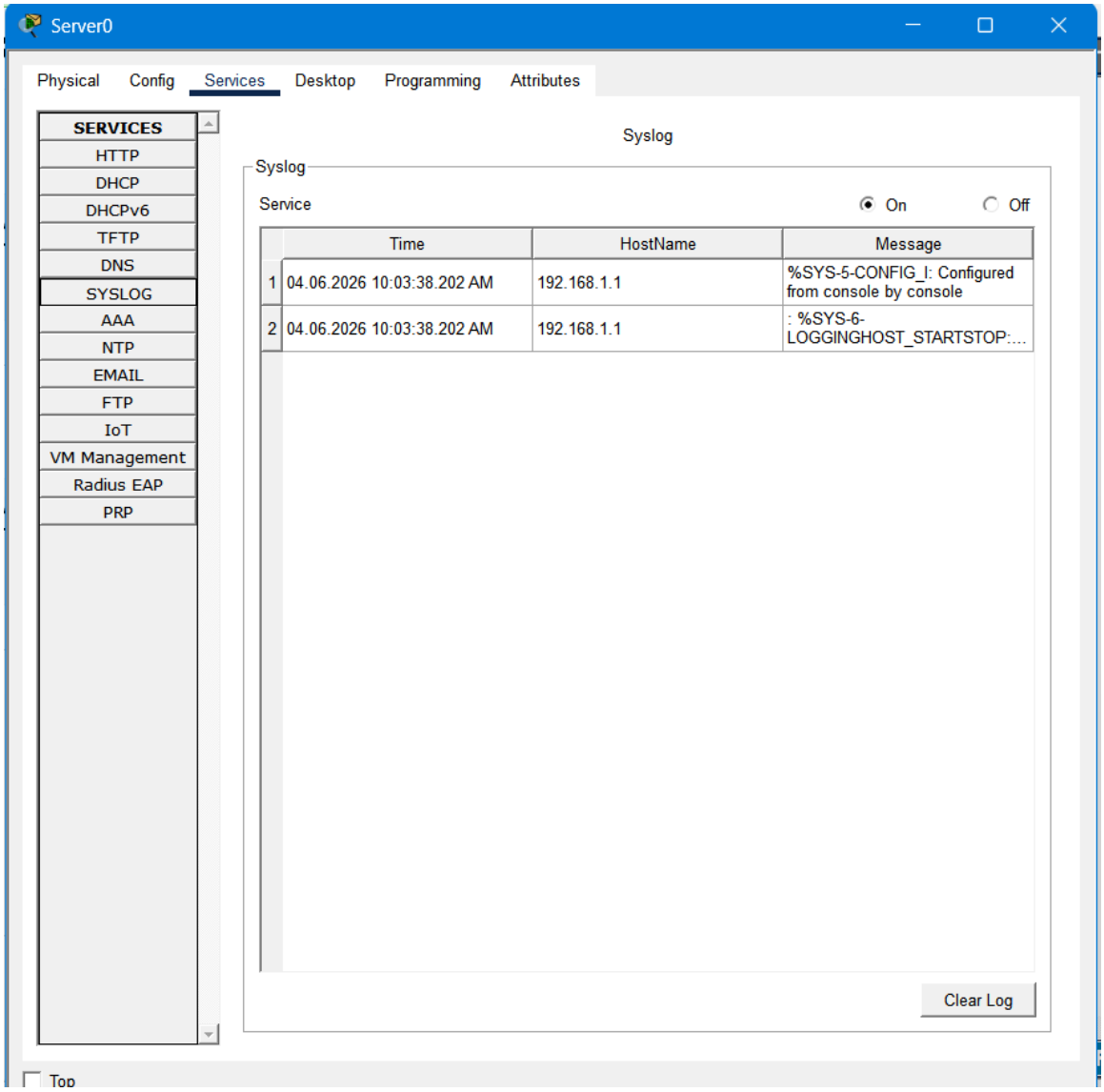
logging host 192.168.1.3

logging trap debugging

Testing and Verification

- Logs were successfully generated on router R1.
- Syslog server (192.168.1.3) received log messages.
- Events such as configuration changes and logging activity were recorded.





Conclusion

Syslog-based monitoring was successfully implemented. Although IPS functionality was not fully available in Packet Tracer, logging provided effective visibility into network events and simulated intrusion detection behavior.

Task	Status
Task 1	✓ Network working
Task 2	✓ SSH + security
Task 3	✓ VPN (conceptual)
Task 4	✓ ACL (access control)
Task 5	✓ Firewall (via ACL workaround)
Task 6	✓ Logging + monitoring

User Credentials

- Username: **admin**
- Password: **admin123**
- Enable secret: **class**
- Console/VTY password: **cisco**